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Can we protect time series data while maintaining accurate forecasts?

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ABSTRACT

We evaluate the usefulness of protected time series by exploring how privacy protection affects forecast accuracy. Using both simulated and real-world time series data sets, we test various privacy methods, including a proposed swapping-based method (*k*-nTS+) designed to maintain time series features, a differentially private method, and an approach based on sharing model weights trained on unprotected data. Based on forecasts from both simple and machine learning models, we find that none of the privacy methods can consistently maintain forecast accuracy at an acceptable level of privacy. We also show that sharing model weights enables accurate forecasts, but accurate forecasts can be used to uncover the identities of protected time series. To overcome these problems, we transform continuous time series into bounded rates to increase the similarities of features, values, and forecasts across time series. This enables our proposed method to produce protected data with a reduction in average forecast accuracy of just 6%. Overall, we acknowledge that except under certain conditions, generating time series with acceptable privacy levels is incompatible with the goal of obtaining accurate forecasts.

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1. Introduction

Consider the two time series in Fig. 1: one with the best of features (a strong positive trend and low spectral entropy), and one with the worst of features (a negligible trend and high spectral entropy). Suppose these series include sensitive data such as monthly household energy consumption, retail transactions, or movements from smart devices (Boone et al., 2019). A data owner could infuse these time series with random noise to limit an adversary from learning sensitive information, but this deprioritizes the usefulness of protected data (which has also been treated as an afterthought by many researchers Blanco-Justicia et al., 2022) and can result in poor forecast accuracy (Gonçalves et al., 2021). To limit the loss of forecast accuracy for protected time series, we propose a feature-based privacy method, *k*-nearest time series+ (*k*-nTS+) swapping, which is designed to maintain time series features useful for forecast accuracy. Instead of infusing random noise, we replace the values of a sensitive time series with values from time series with similar features. This paper addresses whether it is possible to create protected time series with good forecast accuracy.[1]

We pursue three goals at the intersection of data privacy and forecasting. First and most importantly, we evaluate the privacy risk and forecast accuracy of seven forecasting models, ranging from exponential smoothing

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[1] A full reproducibility assessment of the results in this paper could not be performed owing to the substantial computation costs involved. However, all code and data are shared for others to reproduce and build upon the work presented in this paper. See the 'Data availability' section for more information.

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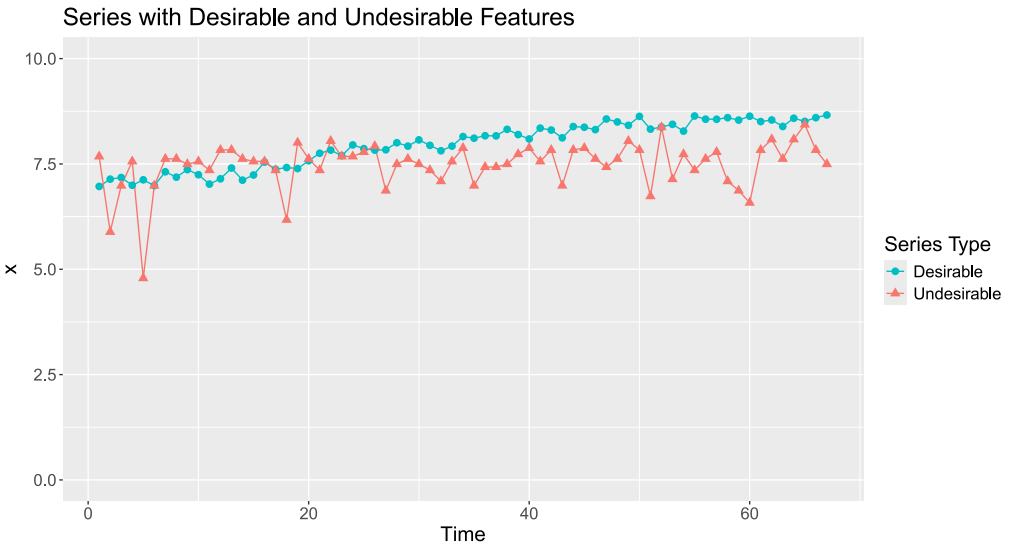


Fig. 1. A tale of two time series: desirable series (blue dots), and undesirable series (red triangles). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

models to a recurrent neural network (RNN). We apply them to simulated time series and the M3 and M4 competition data sets protected using four different privacy methods. We evaluate privacy by measuring the probability of reidentifying protected time series using past protected values (Nin & Torra, 2009) and forecasts. Our results show that none of the privacy methods can generate protected time series with good forecast accuracy at acceptable privacy levels. We find that data that are differentially private (with poor privacy levels of $\epsilon \geq 10$) provide weak protection against reidentification, and that sharing model parameters trained on unprotected data can enable accurate forecasts, but compromises privacy.

Second, we develop a method to share protected time series data with good forecast accuracy (European Parliament and Council of European Union, 2016) under certain scenarios. To the best of our knowledge, this paper is the first to produce a protected time series data set with acceptable reidentification levels and usable forecast accuracy. Importantly, our method enables entire protected time series to be shared with forecasters, rather than model weights or forecasts only, and can accommodate any forecast horizon, accuracy metric, or forecasting model. However, achieving an acceptable tradeoff comes at a cost: time series must be transformed into rates that normalize time series values and increase both cross-series feature and forecast similarity. This transformation enables our method to generate protected time series with a reduction in average forecast accuracy of just 6%, and a reduction in accuracy of 19% when the rate forecasts are transformed back to the original time series scale.

Third, we analyze how time series features change vis-à-vis accuracy and show how various privacy methods change time series features. The results show that the *k*-nTS+ protected data preserve the most accurate feature distributions, although all privacy methods degrade time series features relative to the unprotected data. When important time series features (such as the mean) are maintained, privacy protection increases the randomness of time series, leading to poor forecast accuracy, which is consistent with the findings of Spiliotis et al. (2020). We also incorporate a machine learning-based feature selection method into *k*-nTS+ which improves the performance of the swapping method from 2%–35% depending on the data set. We find that features that are statistically significant predictors of forecast accuracy (Makridakis et al., 2018; Spiliotis et al., 2020) are not necessarily the most useful for swapping time series values.

The rest of the paper is as follows. In Section 2, we review the relevant literature. Section 3 describes the privacy methods and metrics. Section 4 evaluates privacy method performance on simulated time series and the M3 and M4 competition data sets, and Section 5 concludes.

2. Literature review

2.1. Privacy background

Despite the past few years of worldwide data protection legislation, the number of reported data breaches has more than doubled. Fundamental drivers of data breaches include the increased availability of data, parties in data sharing collaborations, and threats from adversaries. Of these threats, it has been estimated that 74% of all data breaches involved human error or deliberate misuse, and 19% of all data breaches were internal involving full-time employees, independent contractors, interns, or other staff.[2] The overarching message is that although privacy can be assured legally and contractually, the data still get out.

To support good privacy practices, data owners may protect time series data by altering the unprotected data values. This approach assumes that the data eventually

[2] See the summary of the 2023 Verizon Data Breach Investigations Report.

get out and protects against the worst-case scenario of a hacker or internal employee breaching the data. Typically, data protection has two goals. The first goal is to limit the ability of an adversary to identify data subjects (identification disclosure, i.e., revealing the true identity of a time series). The second goal is to limit the ability of an adversary to learn sensitive information about data subjects (attribute disclosure, e.g., learning sensitive values after an identification is made). In tandem, these goals limit an adversary's increased knowledge about specific data subjects and have been a primary focus of data privacy researchers (Dwork & Naor, 2010; Hu, 2019; Reiter, 2005).

Time series data are either stored in a single data set (centralized) or spread across multiple data owners and/or data sets (decentralized). In the decentralized scenario (e.g., in the renewable energy sector), secure multi-party computation and decomposition-based methods enable privacy-preserving collaborative forecasting to ensure accurate forecasts while protecting sensitive data (Gonçalves et al., 2021; Sommer et al., 2021). Forecasting sensitive data using data markets (Goncalves et al., 2020) or by sharing forecast model parameters (Goncalves et al., 2021) reduces the availability of sensitive data to potential adversaries but deprives forecasters of useful time series data and may require assumptions of the data generating process. Furthermore, there are privacy risks including potential data breaches with the transfer and storage of time series data with a market operator.

In a centralized scenario, the data owner anonymizes the time series data by directly altering the values to create a protected data set with reduced privacy risk. Using random noise to create differentially private time series is one such alteration, but (Gonçalves et al., 2021) found that the tradeoff between the forecast accuracy of vector autoregressive (VAR) models and differential privacy levels was poor, even for weak levels of differential privacy with $\epsilon = 20$ (it is recommended that ϵ lie in the range $[0.01, ln\ 3]$ (Blanco-Justicia et al., 2022; Dwork, 2011)). Luo et al. (2018) simulated data integrity attacks and found that additive noise reduced forecast accuracy by over 21% when only half the data points were altered. Their results likely understated the reduction in forecast accuracy from privacy methods because only half the data points were altered.

Data records can also be generalized (aggregated or combined) to make every record (or time series) identical to at least $k - 1$ other records (or time series) to improve privacy. For example, daily time series data can be aggregated to weekly time series data (temporal aggregation), or each time series can be averaged with its most similar time series ($k = 2$ anonymity). Nin and Torra (2009) found that $k = 2$ anonymity (weak privacy) resulted in reductions in forecast accuracy for exponential smoothing and regression models. Top and bottom coding can be used to replace the tails of sensitive data distributions with a threshold value, such as $150,000 for income or 10 kilowatts-hours for household energy usage. This limits attribute disclosure risk (i.e., prevents knowledge of specific values within a time series), but may not limit identification disclosure risk (i.e., prevent the identification of an entire time series).

In addition to sharing protected time series data with good forecast accuracy, privacy-enhancing technologies can create a competitive advantage through increased consumer loyalty, trust, and positive performance (Martin & Murphy, 2017). These effects could mitigate damaging outcomes, such as poor stock returns and consumers falsifying information, that arise when consumers feel that their data are vulnerable. Furthermore, replacing sensitive data with protected data could dampen the negative effects of a data breach if the data were to be compromised (Martin et al., 2017). There are also potential benefits from complying with privacy law by using protected time series, such as avoiding the need for consumer consent to reuse data, removing data retention limits, and enabling cross-border data transfers (Arbuckle & El Emam, 2020; European Parliament and Council of European Union, 2016). Using protected data could also help organizations avoid fines such as the $1.3 billion fined against Meta for transferring user data from the EU to the US.[3].

2.2. Time series features

We review the literature on time series features to motivate our time series feature-based privacy method (k-nTS+). Table 1 illustrates a subset of features often examined in the literature along with values of these features from the desirable and undesirable time series in Fig. 1. Several features, such as spectral entropy and the error ACF, have been shown to be important determinants of forecast accuracy (Makridakis et al., 2018). By regressing the mean absolute scaled error (MASE) results from the M4 competition on time series features, Spiliotis et al. (2020) found that increasing the value of the frequency, kurtosis, error ACF, and seasonality of time series improved forecast accuracy, but increasing skewness, Hurst, and spectral entropy degraded forecast accuracy.

Time series features can also be used to improve performance on various forecasting tasks. For example, Bandara et al. (2020) clustered similar time series based on 18 interpretable features, including mean, variance, and seasonality, to improve the accuracy of RNNs between 2% and 11%. Using features as the basis for forecast model selection can improve accuracy, as shown by Qi et al. (2023), who used the strength of trend and seasonality to select exponential smoothing models with better accuracy. Petropoulos and Siemsen (2023) proposed a representativeness metric that selected forecasting models in accordance with the strength of the trend and seasonality of time series, resulting in the best model being selected approximately 3% more often than competing selection methods. The literature also includes methods that use time series features to inform model or forecast combinations that improve the accuracy of forecasts and prediction intervals for the M3 and M4 data sets (Li, Kang, & Li, 2023; Montero-Manso et al., 2020) and can offer lower computational cost compared to top-performing methods in the M4 competition (Talagala et al., 2022). Interestingly, using the diversity of forecasts as a single feature achieves

[3] E.U.slapsMetawithrecord$1.3billionfinefordataprivacyviolations.

Table 1
Time series feature descriptions, ranges, and values from Fig. 1.

Feature	Description	Value range	Desirable series	Undesirable series
Spectral entropy	Signal-to-noise ratio of a time series.	[0, 1]	0.07	1.00
Hurst	Long-range dependence (self-similarity) of a time series.	[0, 1]	0.99	0.50
Skewness	Symmetry of the distribution of time series values.	$(-\infty, \infty)$	−0.41	−0.57
Kurtosis	Weight of the tails of the distribution of time series values.	$(-\infty, \infty)$	−1.24	1.16
Error ACF	First autocorrelation coefficient of the error component of a decomposed series.	[−1, 1]	−0.09	−0.19
Trend	Strength of the trend.	[0, 1]	0.97	0.12
Seasonality	Strength of the seasonality.	[0, 1]	0.16	0.23
Mean	Mean of a time series.	$[0, \infty)$	7.96	7.01
Variance	Variance of a time series.	$[0, \infty)$	0.29	0.65

similar or better forecast accuracy compared to using dozens of features from historical data (Kang et al., 2022; Li, Kang, Petropoulos, & Li, 2023). Overall, these findings suggest that a time series feature-based privacy method will give us the best chance at maintaining the forecast accuracy of protected time series.

3. Privacy methods and metrics

In this section, we outline the various privacy methods that we apply to the simulated time series and the M3 and M4 competition data sets in Section 4, and the metrics used to assess the privacy risk of the protected time series.

3.1. Differential privacy

We follow the interpretation and implementation of differential privacy from (Gonçalves et al., 2021). A mechanism M satisfies ϵ-differential privacy by guaranteeing that, for every output x' of M and every pair of series x_j and x_i which differ on at most one observation:

$$\Pr\Big(M(x_j) = x'\Big) \leq \exp(\epsilon)\Pr\Big(M(x_i) = x'\Big). \quad (1)$$

A differentially private time series can be created using a randomized mechanism $M(x_j) = x_j + \eta$ that adds a vector of random noise values, each of which is drawn from a Laplace distribution with scale parameter $\Delta f_1/\epsilon$, to an original time series x_j. The sensitivity Δf_1 is determined as the maximum absolute difference between two time series x_j and x_i, which differ in at most one observation, where $\Delta f_1 = \max \|x_i - x_j\|_1$. The privacy parameter ϵ determines the level of privacy, which decreases as ϵ increases.

3.2. Additive noise

Additive noise adds a normally distributed random number with mean zero and standard deviation σ to each value in a time series x. Protected values can be written $P_t = A_t + r$, where $r \sim N\left(0, \sigma^2\right)$ and $\sigma = s\sigma_{x_j}$. The protection parameter s denotes the number of standard deviations of x_j, and privacy increases as s increases.

3.3. k-nTS

We propose a matrix-based k-nTS (k-nearest time series) swapping method, where the data owner releases a set of protected time series $X' = [x'_1, \ldots, x'_J]$, where $x'_j = \left(P_{j,1}, \ldots, P_{j,T}\right)^T$ for $j = 1, \ldots, J$. The k-nTS method uses two parameters: k (the number of neighbors to consider for swapping), and m (the swapping window length). To create a protected series x'_j, the k-nTS swapping method finds the k most similar time series to the first window of m values of x_j, where similarity is based on time series features. For each of the time periods $t = 1, \ldots, m$, one of the k similar series is selected at random and the value in time t in x_j is replaced with the value from time t from the selected series. The values in time periods $t = m + 1, \ldots, T$ in x_j are replaced in the same manner, where the window is rolled forward and the features and neighbors are recalculated for each successive period. We give a detailed description of the k-nTS privacy method in Section 1 of the appendix, including equations for the feature distance matrix, the option of weighting the distance calculation based on the importance of different time series features, and the time complexity of the k-nTS swapping algorithm (Algorithm 1).

3.4. k-nTS+

The k-nTS+ swapping method adds a feature selection process to k-nTS swapping which selects features that

are good predictors of forecast accuracy. The data owner can provide a large selection of time series features to the method. However, using all of the features for the basis of swapping would significantly increase the dimensionality and reduce the efficiency of the swapping process. To address this problem, a random forest-based recursive feature elimination (RFE) algorithm is applied to features initially selected by RReliefF (Robnik-Šikonja & Kononenko, 2003). Prior work has shown that random forest-based RFE is efficient when applied to sets of highly correlated features (Gregorutti et al., 2017), ensuring that the method selects a small set of features that predict forecast accuracy well. The detailed algorithms for feature selection (Algorithm 2) and k-nTS+ (Algorithm 3) are given in Section 2 in the appendix; we provide an overview in this section.

The feature selection methodology proceeds as follows. Let f_j^κ denote one of the $\mathcal{K}$ nearest neighbor feature vectors to f_j, the feature vector of series x_j, where $\mathcal{K}$ is the number of nearest neighbors considered by RReliefF. Let ϵ_j^κ and ϵ_j denote the forecast errors for the corresponding time series. Let π_ϵ and π_n denote the events that series x_j and x_j^κ have different forecast errors and different values for the nth feature, respectively, conditional on being nearest neighbors. The RReliefF weight for the nth feature approximates the difference in conditional probabilities:

$$\omega_n = p\left(\pi_n \middle| \pi_\epsilon\right) - p\left(\pi_n \middle| \pi_\epsilon^c\right). \tag{2}$$

This weight approximates the difference between the probability that the nth feature discriminates between series with different forecast errors, and the probability that the nth feature discriminates between series with the same forecast error. In the first stage of the feature selection method we eliminate all features that are poor predictors of differences in forecast error (i.e., in case of $\omega_n \leq 0$).

The second stage of the feature selection method utilizes random forest-based RFE. First, we train a random forest model to predict the forecast errors from a given forecasting model using a set of time series features and compute the mean absolute error (MAE) on the out-of-bag (OOB) predictions using that set of features. Next, we compute the permutation-based feature importance, i.e., the change in OOB MAE from permuting the values of a given feature. The least important feature is dropped from the model, and the model training and feature elimination process is performed recursively until one feature remains. Each feature is then assigned a rank based on the reverse of the elimination order. This process is performed N^{rfe} times. Different permutations of the data may result in different importance measures across RFE iterations. Thus, we compute s^*, the number of features that resulted in the lowest average OOB MAE across all RFE iterations, and select the s^* features with the best (lowest) average rank across all RFE iterations.

Fig. 2 illustrates the full k-nTS+ protection process, which can be used collaboratively between a data owner and the forecaster. The forecaster specifies their preferred forecasting model $\mathcal{M}$. The data owner generates protected versions of the original data set up through time period $T-h$ using baseline privacy methods with varying levels of privacy strength. The data owner computes the forecast errors from using the model $\mathcal{M}$ to forecast time periods $T-h+1$ through T for each version of the data. These errors are the target variable in the feature selection process, which is performed on a feature matrix $\mathcal{C}$ containing the combined feature values from the original and baseline protected data sets. We note that the feature values should be normalized *after* they are combined in $\mathcal{C}$. The selected features F^* are then used as the basis for swapping to create the protected data set X' from the original data set up through time period T. The data owner can perform the swapping process over successive time periods conditional on the selected features to release protected data for times $T+1$ and beyond, or may choose to perform the feature selection process again for any time period.

3.5. Sharing model weights

Instead of sharing protected data for forecasting, an alternative is for the data owner to share pre-trained models with the forecaster, as seen in collaborative scenarios (Goncalves et al., 2021). We simulate a scenario where the data owner shares VAR model parameters from a model trained on the unprotected data with the forecaster. For example, suppose the data owner trains a VAR model with $l=1$ lags on $J=5$ differenced unprotected time series over time periods $t=2,\ldots,T$. Using the VAR model and the first $l \times J$ differenced time series values, the data owner simulates $V=5000$ potential future paths for each of the J series for the next $l+1$ time periods, and takes the average of the potential values in each period for each time series. These values, along with the model intercepts, lag coefficients, and the covariance matrix of the model residuals, are shared with the forecaster. To generate forecasts, the forecaster continues the simulation process described above through time T, and generates forecasts using the model weights and $l \times J$ most recent values of the simulated series. Overall, this approach enables the forecaster to obtain a data set (of simulated time series) and model weights trained on the unprotected data without sharing confidential data.

3.6. Identification and forecast disclosure risk

We assess the ability of each privacy method to protect against identification disclosure, which occurs when an adversary correctly predicts the identity of a protected time series. While preventing (or limiting the success of) identification disclosure attacks is not a theoretical privacy guarantee, it is an interpretable and reasonable standard of privacy that can enable data owners to achieve a targeted level of privacy protection while maintaining useful data (Bale et al., 2024).[4] Each protected data set consists of the protected series along with a pseudo-identifier (PID), i.e., $X' = \left[\left(PID_1, x_1'\right), \left(PID_2, x_2'\right), \ldots \left(PID_J, x_J'\right)\right]^T$. Identification disclosure occurs if an adversary (or forecaster) correctly predicts the identity of one

[4] See the Article 29 Working Party Guidelines 05/2014.

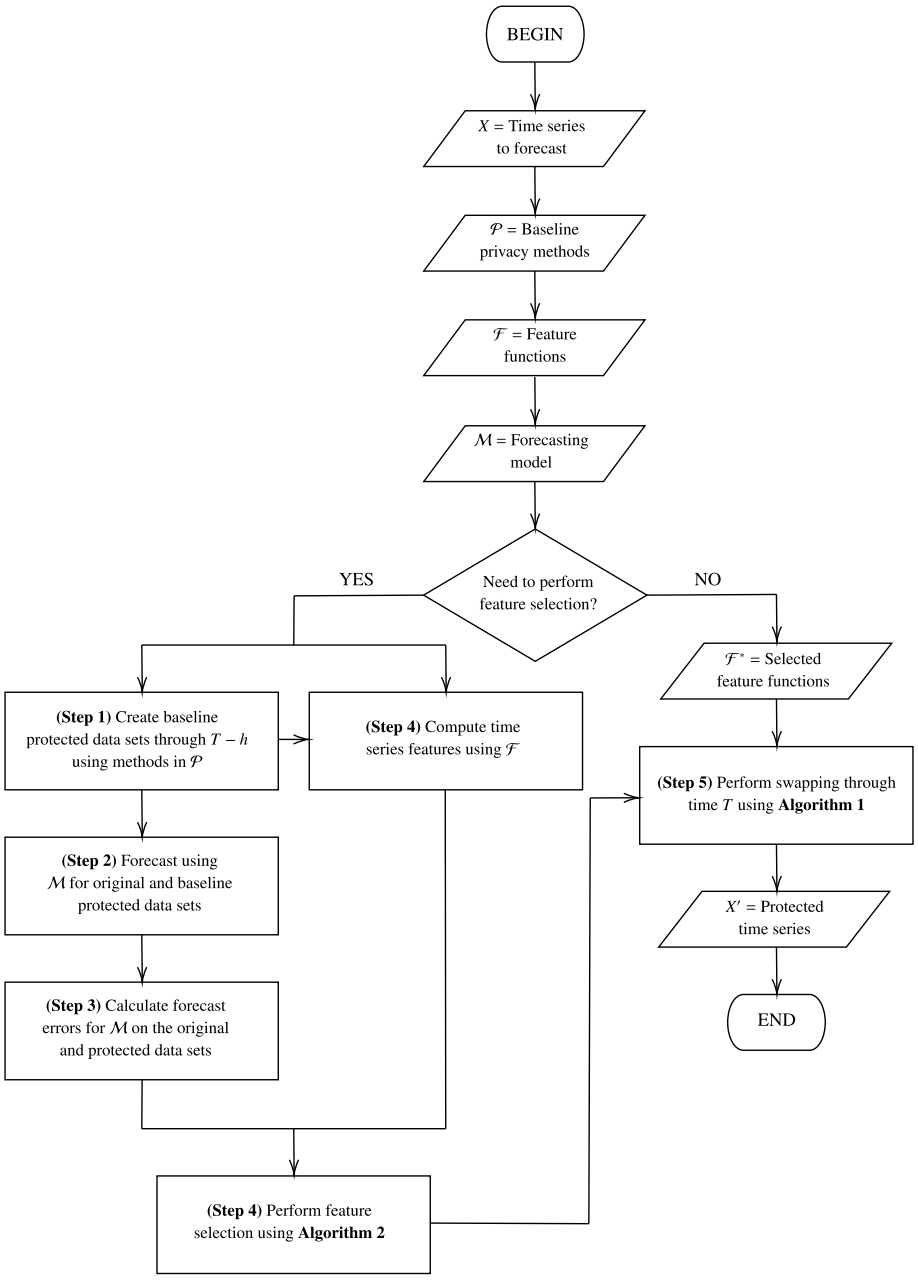


Fig. 2. Flowchart of the k-nTS+ data protection process.

or more of the protected time series based on the protected time series and some outside information the adversary possesses. For example, identification disclosure occurs when an adversary correctly states, 'series Y comes from the monthly sales of the Roseville, Minnesota Target store.' An acceptable level of identification disclosure risk depends on the context in which the protected data will be released and the preferences of the data owner. One example is the European Medicines Agency (EMA), which has set a maximum acceptable identification probability of 9% (European Medicines Agency, 2017).

We assess an identification disclosure attack where an adversary uses unprotected time series values to identify the protected time series. We perform $S = 20$ simulations of this privacy attack in which we sample ten sequential values from each unprotected time series and treat these as external information available to the adversary. The adversary predicts the identity of each protected series based on the unprotected values that are closest to the protected values from the same time periods.

The metric we use is the identification disclosure risk for time series (Nin & Torra, 2009), $\bar{P}$, or the average proportion of the J time series which are correctly identified

across the S simulated privacy attacks:

$$\bar{P} = \frac{1}{J \times S} \sum_{s=1}^{S} \sum_{i=1}^{J} I\left(\hat{M}_i^s = j^*\right), \tag{3}$$

where $\hat{M}_i^s$ is the adversary's prediction of the identity of the ith protected time series. We use $I(*)$ to denote the indicator function which is equal to one when identification disclosure occurs, i.e., when the predicted identity is equal to the true identity j^* (and zero, otherwise). We refer the reader to Section 11 of the appendix for added mathematical details.

In addition to identification disclosure risk, we measure the proportion of time series which can be correctly identified based on their forecasts, which we term the forecast disclosure risk. The attack is similar to the one described above, except instead of using past unprotected values to identify protected series, the adversary compares unprotected values from future time periods to the forecasts from protected data. Let $\hat{X} = [(PID_1, \hat{A}_{1,T+1}), (PID_2, \hat{A}_{2,T+1}), \ldots, (PID_J, \hat{A}_{J,T+1})]$ denote the set of forecasts and pseudo-identifiers for the protected time series, where $\hat{A}_{j,T+1}$ is the forecast for the jth series for time $T+1$. Suppose the adversary observes the actual unprotected values for time $T+1$ as external information and compares these values to the forecasts. The adversary predicts the identity of the ith protected time series based on which forecast is most similar to the jth unprotected value. Forecast disclosure occurs when an identification is made correctly, and we measure forecast disclosure using $\bar{P}^f$:

$$\bar{P}^f = \frac{1}{J} \sum_{i=1}^{J} I\left(\hat{M}_i^f = j^*\right), \tag{4}$$

where $\hat{M}_i^f$ is the adversary's prediction of the identity of the ith time series based on the forecasts. Although identification and forecast disclosure are only two of many possible attacks, successfully reidentifying a large portion of time series (e.g., $> 9\%$) indicates that a time series data set is not sufficiently protected.

The attacks described in this section do not capture all of the ways identification disclosure could occur. For example, the values of features such as the mean and variance could uniquely identify time series, and it may be better for privacy if the values of these features are not *exactly* preserved in the protected data. Overall, the metric in (3) provides a reasonable lower bound on the identification disclosure risk associated with protected time series.

4. Empirical application

As mentioned above, we apply the privacy methods to simulated time series and the M3 and M4 competition data sets. For brevity, Section 4.1 gives a summary of the results from the simulated time series, and detailed results are given in Section 3 of the appendix. Section 4.2 describes detailed results from the M3 competition data set and summarizes the M4 results, with more detailed M4 results given in Section 5 of the appendix.

4.1. Simulation: A tale of two (sets of) time series

We first illustrate the performance of the privacy methods using two sets of 10 time series each, with 'desirable' (low spectral entropy and high autocorrelation) and 'undesirable' (high spectral entropy and low autocorrelation) features, respectively. The series are simulated using the GRATIS (GenerRAting TIme Series) R package (Kang et al., 2020). Each of the simulated time series is re-scaled using a mean and standard deviation sampled from uniform distributions over the ranges [2000, 12000] and [50, 500], respectively, and the series are depicted in Fig. 3.

We create protected data sets using differential privacy (for 39 values of ϵ ranging from 20 to 0.01), additive noise (for 39 values of s ranging from 0.08 to 30.00), and k-nTS+, (for a window length of $m = 25$ and $k = 2, 3, 4, 5, 6, 7, 8, 9$). The baseline protected data sets for k-nTS+ are created using differentially private noise, and we include all features which could help improve accuracy in the swapping process and which are easily computable using the 'tsfeatures' package in R (Hyndman et al., 2023). We refer the reader to Yang and Hyndman (2023) for a detailed explanation of time series features and provide further mathematical detail on several time series features in Section 12 in the appendix. For the AN, DP, and k-nTS+ protected data, we generate forecasts using simple exponential smoothing (SES) and double exponential smoothing (DES) forecasting models. For sharing model weights, we use two VAR models of lag order one on each data set trained on five time series each, and compare the results to forecasting using VAR models trained on the k-nTS+ protected data. For each protected data set, we perform the privacy simulation discussed in Section 3.6 and account for the randomness of privacy protection by reporting average results across 25 protected data sets for each privacy parameter.

4.1.1. Simulation results summary

A detailed description of the simulation results is given in Section 3 of the appendix, including results for larger sets of simulated time series. Fig. 4 shows the average $\bar{P}$ (top row) and the average MAE (bottom row) for each set of time series across the 25 simulations for each privacy parameter for AN, DP, and k-nTS+. In short, while k-nTS+ offered a significant improvement in the tradeoff between privacy and forecast accuracy relative to additive noise and differential privacy, the forecasts based on the k-nTS+ protected data were still unusable due to poor accuracy. Notably, the VAR-simulated series (results shown in Table 2) offered significantly better forecast accuracy (when the models did not diverge) than the other privacy methods, but this accuracy improvement came at the cost of privacy; on average, over 60% of the VAR-simulated series could be correctly identified using the identification disclosure attack, and at least 20% could be identified based only on one forecast. k-nTS+ provided much stronger privacy protection with identification and forecast disclosure risk estimates at or below the 9% threshold.

Our results show that achieving good forecast accuracy at an acceptable level of privacy is not possible on small sets of time series with unique means and variances.

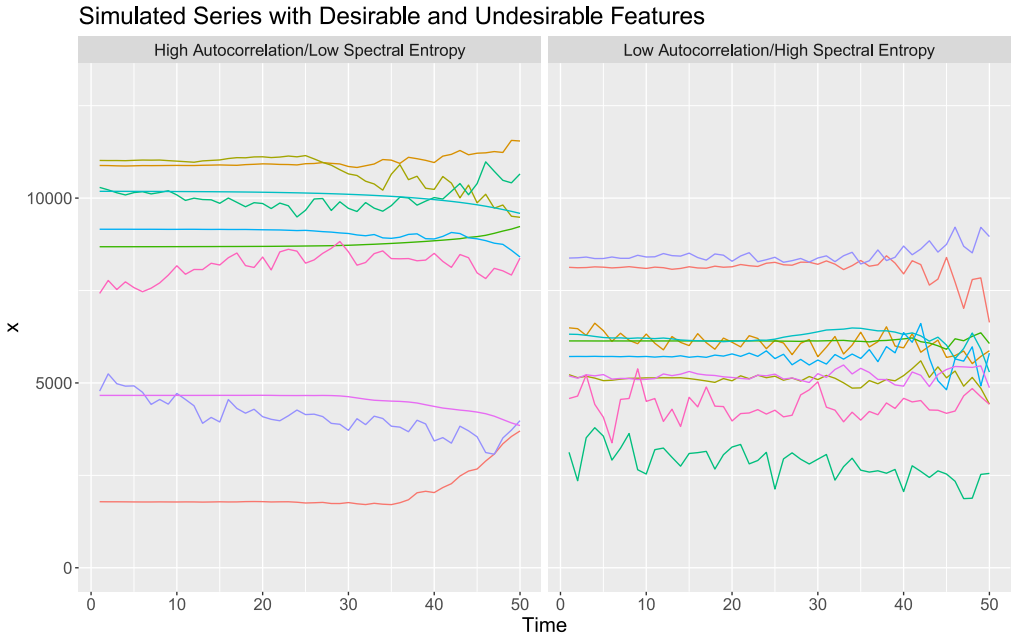


Fig. 3. A tale of two (sets) of time series. The sets of 10 series with desirable and undesirable features are shown in the left and right plots, respectively.

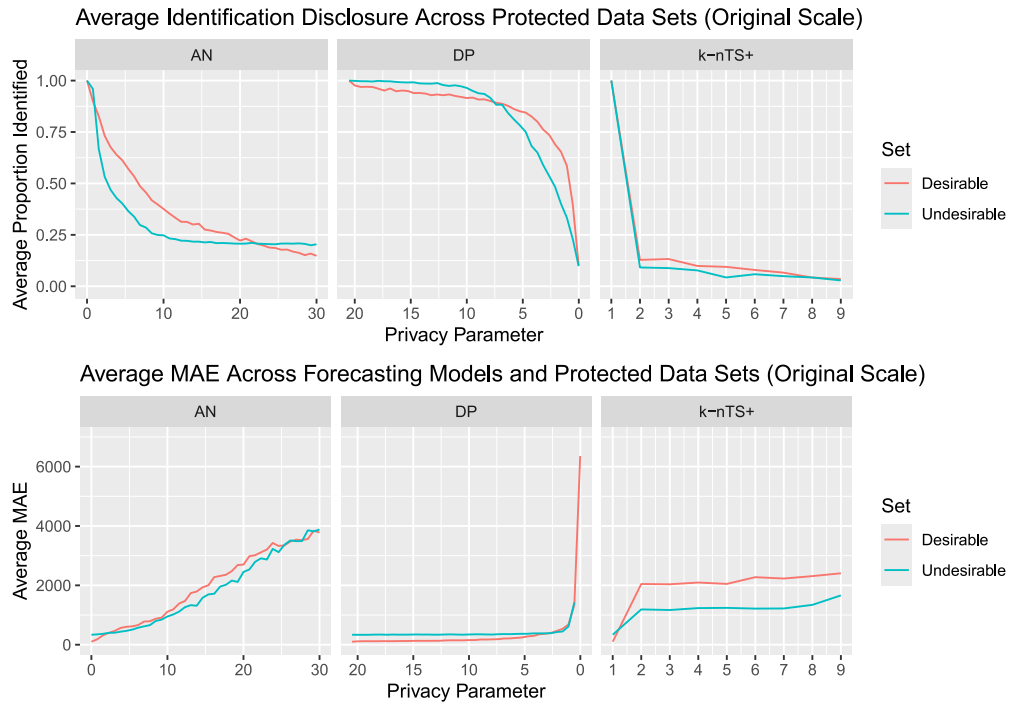


Fig. 4. Average identification disclosure risk (top row) and average MAE (bottom row) for all privacy parameters for additive noise, differential privacy, and *k*-nTS+ applied to sets of 10 time series with desirable or undesirable features.

These time series had very different values (and features) and were easily identified, which is consistent with privacy results on non-time series data—that a small set of quasi-identifying attributes is often sufficient to identify large portions of data sets (Sweeney, 2002). DP, AN, *k*-nTS+, and the VAR-simulated approach were not able to produce protected versions of these series with acceptable privacy protection *and* forecast accuracy. Either the time series values (and features) are changed drastically to achieve acceptable privacy protection with unacceptable forecast accuracy (e.g., under *k*-nTS+), or values and features are preserved with little to no privacy (e.g., the VAR-simulated series).

To create conditions where protected data can be released with good forecast accuracy, we transform the simulated time series into rates prior to performing the data protection and forecasting tasks. Rates are preferred to original values for privacy reasons, since they are bounded (Abowd et al., 2013), which relates to the global sensitivity of differential privacy (Dwork, 2011). The transformation makes the series less identifiable by normalizing values across series with different means and other features. This also reduces the identifiability of functions of time series values, such as a mean forecast. In addition, the ability of our proposed *k*-nTS+ method to swap time series values with minimal detriment to forecast accuracy will improve, since the swappable values for any given time period will be on a similar scale to the replaced value. We perform the transformation using the following formula:

$$r_t = (A_t - A_{t-1}) / \bar{A}_t, \tag{5}$$

where $\bar{A}_t = (A_t + A_{t-1})/2$. We set $r_1 = 0$ to begin the conversion. Assuming that the original time series values are non-negative, the resulting rates will be bounded to $[-2, 2]$. If sequential zeros occur in the original data, i.e., if $A_t = A_{t-1} = 0$, the rate in (5) will be undefined and will have to be manually set to zero.

Results from applying the AN and DP methods to the simulated rate series can be found in Table 1 of the appendix. In Table 2 we compare the VAR forecast accuracy (measured as the percent change in average MAE relative to the original data) and privacy results based on the *k*-nTS+ protected data sets and the VAR-simulated approach directly. In general, we see improvements in the privacy risk and VAR model forecast accuracy for both methods relative to using the original data. The identification disclosure risk for the VAR-simulated series reduces to random guessing while accuracy is maintained (undesirable series) or even improves (desirable series). However, the forecast disclosure risk for the VAR-simulated approach remains high for the desirable series. In comparison, the forecasts from the *k*-nTS+ protected data offered accuracy improvements from around 4%–19% while keeping average forecast disclosure well under random guessing.

4.2. Competition data application

In this section, we assess the performance of the privacy methods on time series from the M3 and M4 competitions, which have features that are representative of real-world data (Spiliotis et al., 2020). The organizers of the early M competitions did not disclose the true identity of the competition time series (Makridakis & Hibon, 2000). This provides a natural connection to privacy because we can compute the identification and forecast disclosure risk of each protected time series. Motivated by the simulation results in Section 4.1, we analyze forecast accuracy and privacy when protecting the original M3 and M4 data sets and after converting the M3 and M4 data sets into rates. For brevity, we focus on the M3 results in this section and include the M4 results in Section 5 of the appendix.

4.2.1. Competition data implementation details

The data owner protects every time series value from time period 1 to T using differential privacy for $\epsilon = 20, 10, 4.6, 1, 0.1$, additive noise for $s = 0.25, 0.5, 1.0, 1.5, 2.0$, and *k*-nTS and *k*-nTS+ for $k = 3, 5, 7, 10, 15$. To perform feature selection for *k*-nTS+, the data owner creates protected versions of the data using both AN and DP from time period 1 to $T - 1$ for the above parameter values and performs feature selection and data protection as described in Sections 3.3 and 3.4 using the features described in Section 4.1. To show that *k*-nTS+ can accommodate multivariate time series data, we include the eigenvectors corresponding to the first five principal components of the time series cross-correlation matrix as additional features. We use $m = 25$ as the window length for the monthly data, and $m = 12$ for other frequencies, as $m \geq 12$ is required for the computation of some of the time series features such as the sum of the first 10 coefficients of the ACF of the twice-differenced time series (second difference ACF10). To demonstrate the uplift in forecast accuracy from using the feature selection method, we compare the results from the *k*-nTS+ protected data to data protected using *k*-nTS with features selected based on the literature review in Section 2.2. We select the mean, variance, spectral entropy, *Hurst*, skewness, kurtosis, error ACF, trend, and seasonality, since these features were used to improve the forecast accuracy of RNNs (Bandara et al., 2020) and/or were statistically significant predictors of forecast accuracy on the M4 competition data (Spiliotis et al., 2020). We omit stability and non-linearity, since these features were not significant predictors of forecast accuracy after controlling for the previously mentioned features (Spiliotis et al., 2020), and we omit the frequency, because none of the privacy methods we consider change the frequency of the original data.

The protected time series are given to forecasters to produce one-step-ahead forecasts for time $T + 1$. For all privacy methods, one-step-ahead forecasts for time $T + 1$ are generated using forecasting models implemented in R and Python and listed in Table 3. Similar to the M competitions, all reported forecast accuracy and standard deviation results are derived from comparing the forecasts for $T + 1$ to the actual data from $T + 1$. For the rate versions of the series, we examine the accuracy of the rate forecasts compared to the actual rate from time $T + 1$, and the rate forecast transformed back to the original time series scale compared to the actual original value from

Table 2
A tale of two (sets of) time series: Comparison of identification ($\bar{P}$) and forecast ($\bar{P}^f$) disclosure risk and VAR model forecast accuracy (percent change in MAE relative to the unprotected data) under k-nTS+ and the VAR-simulated approach for the protected versions of 10 simulated series with desirable and undesirable features in original and rate form.

Method	Data set	Parameter	$\bar{P}$	$\bar{P}^f$	% Change MAE
k-nTS+	Desirable	$k = 4$	8.44%	9.20%	4141.39%
	Desirable (rate)	$k = 2$	5.11%	3.60%	−4.51%
	Undesirable	$k = 7$	5.98%	6.40%	420.13%
	Undesirable (rate)	$k = 2$	2.34%	7.60%	−19.26%
VAR-simulated	Desirable	–	61.50%	20.00%	3782.08%
	Desirable (rate)	–	10.00%	20.00%	−11.26%
	Undesirable	–	64.00%	60.00%	−0.01%
	Undesirable (rate)	–	10.10%	10.00%	1.70%

Table 3
Univariate and multivariate forecasting models applied to the M3 and M4 competition data sets.

Model	Variant
SES	–
DES	Additive trend
TES	Additive trend/seasonality
Auto-ARIMA	Seasonal
VAR	–
LGBM	–
RNN	–

$T + 1$. The light gradient-boosting machine (LGBM), RNN, and VAR models are trained separately on the subsets of time series with equal lengths. We perform minimal data pre-processing and use the standard settings in the off-the-shelf packages. Full implementation details can be found in Section 10 of the appendix.

Finally, we assume that the forecaster may be an adversary attempting to identify an unprotected time series using the protected time series. We take a conservative approach when calculating disclosure risks and assume that the adversary knows the length of each time series, which makes identification easier. Thus, we separate the privacy analysis and protection into the groups of time series with equal lengths within each of the M competition data sets. Reported privacy results are derived from calculating the identification disclosure risk using the protected data from time period 1 to T, and from calculating the forecast disclosure risk using the actual data and forecasts for time period $T + 1$. We note that in order to test our proposed method with the desired values of k, we exclude any series that do not have the same length as at least 15 other series. This left us with 2363 of the 3003 M3 competition series, where approximately 1.3% of these protected series would be correctly identified from random guessing.

The rest of the empirical application is outlined as follows. Sections 4.2.2 and 4.2.3 present the privacy and forecast accuracy results, Sections 4.2.4 and 4.2.5 analyze the results of the feature selection methodology and examine how time series features change under each privacy method, and Section 4.2.6 discusses the computational cost of the proposed k-nTS+ privacy method.

4.2.2. Privacy and accuracy results

Table 4 displays the percentage change in the MAE of one-step-ahead point forecasts and the disclosure risk metrics $\bar{P}$ and $\bar{P}^f$ across all models and M3 series.[5] For each privacy method, we display the two privacy parameters that had identification disclosure risk measures closest to the 9% threshold.

The results show an inverse relationship between forecast accuracy and the strength of privacy protection. However, on average for time series on the original scale, none of the methods offer acceptable protection against identification and forecast disclosure while maintaining an acceptable level of forecast accuracy, which is consistent with the results from the simulation in Section 4.1. While not shown in this table, even small amounts of random noise that offer little privacy protection (additive noise with $s = 0.25$ and differential privacy with $\epsilon = 20$) degrade forecast accuracy by approximately 30%. On the other hand, k-nTS and k-nTS+ offer acceptable protection against reidentification at any value of k but produce significant reductions in forecast accuracy of over 100%. For a comparable level of privacy (about 3%–8% identification disclosure risk) the k-nTS+ ($k = 3$) protected series had approximately 170% better forecast accuracy than under additive noise ($s = 2.00$) and 578% better accuracy than under differential privacy ($\epsilon = 1.00$). Furthermore, k-nTS+ had over 30% better forecast accuracy than k-nTS with manually selected features, which shows the uplift from the proposed feature selection methodology.[6]

[5] We also measure the percentage change in the MAE for the time series with the 50% largest and smallest magnitudes (defined by the median of the true values for the forecasted time period $T + 1$). A portion of these results are shown in Section 4 in the appendix. For AN and DP, we observe lower percentage increases in the MAE for the smaller magnitude series than the larger magnitude series. However, conclusions drawn from these values are consistent with what is discussed in the rest of this section regardless of the magnitude of the time series—that none of the privacy methods produce protected time series on their original scale with usable forecast accuracy at acceptable privacy levels. For k-nTS and k-nTS+, we observe little difference (at most 16%) between the percent increase in forecast error for the large vs. small magnitude time series.

[6] All accuracy results exclude the application of the VAR model to the yearly micro-subset of the M3 data, due to the model producing explosive forecasts that skewed the overall average results. Similarly, the averages for additive noise with $s \in \{1.5, 2\}$ and for differential privacy with $\epsilon \in \{0.1, 1.0, 4.6\}$ exclude results from the VAR model applied to various subsets of the M3 data that produced MAE values over 100,000. Even with the large amount of random noise infused from these privacy methods, the VAR model did not decrease the magnitude of coefficients to smooth out the noise, resulting in extremely poor forecast accuracy. This is especially problematic if, for example, an

Table 4
Identification disclosure risk and forecast accuracy for the unprotected and protected original and rate M3 data sets.

Privacy method	Parameter	$\bar{P}$		$\bar{P}^f$		Accuracy (% change MAE)		Accuracy (% change MAE) original scale
		Original	*Rate*	*Original*	*Rate*	*Original*	*Rate*	*Rate*
Unprotected	–	98.4%	98.4%	18.4%	2.5%	0%	0%	0%
Additive noise	$s = 1.5$	11.1%	7.3%	5.9%	1.9%	218.7%	23.7%	80.0%
	$s = 2$	8.0%	3.4%	4.8%	1.7%	298.7%	35.6%	131.3%
Differential privacy	$\epsilon = 4.6$	15.7%	10.3%	6.9%	2.0%	209.4%	29.6%	826.1%
	$\epsilon = 1$	3.7%	1.4%	2.6%	1.7%	705.9%	186.5%	–
k-nTS	$k = 3$	3.5%	4.5%	2.6%	1.4%	163.0%	8.1%	21.4%
	$k = 5$	3.7%	4.2%	2.3%	1.5%	178.5%	8.8%	23.5%
k-nTS+	$k = 3$	5.2%	7.8%	3.0%	1.5%	127.9%	6.1%	19.3%
	$k = 5$	4.7%	6.5%	2.9%	1.4%	143.6%	6.7%	21.3%
k-nTS+ ($k = 3$)	$M = 1$	14.3%	30.6%	5.2%	1.6%	74.4%	6.5%	17.4%
	$M = 1.5$	8.2%	15.1%	4.3%	1.6%	92.3%	7.4%	18.2%

We attempt to improve the forecast accuracy of the *k*-nTS+ ($k = 3$) protected data by implementing an additional post-processing step. Let $B_j \geq \left|A_{j,t} - P_{j,t}\right|$ denote a non-negative constant that bounds the absolute difference between the original and protected time series values in time t for the jth series. In this case, $P_{j,t}$ is the value that was swapped into time period t during the *k*-nTS+ protection process. In related work, Lee and Schneider (2023) derive robust bounds on the absolute change in additive exponential smoothing-based forecasts based on protected data.

We define $B_j = M\sigma_{x_j}$, where M is a non-negative constant and σ_{x_j} denotes the standard deviation of time series x_j. Let $P^B_{j,t}$ denote the protected value for series j in time t after bounding the absolute difference between the unprotected and protected values in series j, which is defined as follows:

$$P^B_{j,t} = \begin{cases} P_{j,t}, & \text{if } \left|A_{j,t} - P_{j,t}\right| \leq B \\ A_{j,t} - B_j, & \text{if } \left|A_{j,t} - P_{j,t}\right| > B \text{ and } A_{j,t} > P_{j,t} \\ A_{j,t} + B_j, & \text{if } \left|A_{j,t} - P_{j,t}\right| > B \text{ and } A_{j,t} < P_{j,t}. \end{cases} \tag{6}$$

The parameter M can be used to control the tradeoff between privacy and forecast accuracy. Smaller (larger) values of M will restrict the protected values $P^B_{j,t}$ to a tighter (looser) interval around the unprotected values $A_{j,t}$, which will increase (reduce) forecast accuracy and increase (reduce) disclosure risks. This post-processing is useful in cases where *k*-nTS+ produces protected data with disclosure risks below the 9% threshold. That is, there is headroom to increase forecast accuracy at the expense of increased disclosure risks while remaining below the required threshold. This post-processing also does not incur significant computational cost because for a given value of k, we can improve forecast accuracy without repeating the swapping process.

We apply the bounding step to the *k*-nTS+ ($k = 3$) protected data set and the results are shown in the last two rows of Table 4. We tested values of $M = 0.5, 1.0, 1.5$. Using $M = 1.5$ improved forecast accuracy by about 35% relative to the *k*-nTS+ ($k = 3$) protected data and increased identification disclosure risk to only 8.2%, but the forecasts are still unusable in this scenario. While using $M = 1.0$ further improves forecast accuracy, it pushes the identification disclosure risk to over 14%. While the bounding improves forecast accuracy at acceptable levels of identification risk, it does not change our overall conclusion that none of the methods produce protected data with acceptable levels of both privacy risk and forecast accuracy.

Table 4 also displays privacy and forecast accuracy results for the M3 rate data. Since the data owner possesses the original time series data, the forecaster can share rate forecasts with the data owner who can transform the rate forecasts back to the scale of the original time series. Let $\hat{r}_{T+1}$ denote the forecasted rate for time $T + 1$. A forecast for time $T + 1$ that matches the scale of the original data can be obtained using $\hat{r}_{T+1}$ and the original data point A_T from time T (which is known by the data owner):

$$\hat{A}_{T+1} = A_T \frac{1 + 0.5\hat{r}_{T+1}}{1 - 0.5\hat{r}_{T+1}}. \tag{7}$$

Column six of Table 4 contains the percent change in MAE for these transformed rate forecasts relative to the MAE of the forecasts based on unprotected data. Under the *k*-nTS and *k*-nTS+ privacy methods, these transformed rate forecasts are usable, offering only 19%–24% less accuracy than the forecasts from the original data. These forecasts are significantly more accurate than any of the forecasts from the protected original data shown in column five of Table 4. We do not include the results for transformed rate forecasts based on differential privacy ($\epsilon = 1$) because many of the forecasts approached infinity and resulted in poor forecast accuracy.

The protected rate data sets consistently exhibit smaller percentage increases in forecast error across all privacy methods and parameters. For *k*-nTS and *k*-nTS+, identification disclosure risk is capped at 8%, with forecast accuracy reductions of approximately 9% or less, which is a better tradeoff than under AN and DP. The post-processing bounding method proved ineffective for *k*-nTS+ ($k = 3$) protected rate data, as identification disclosure risk was pushed to unacceptable levels with no

extreme outlier occurs in a series at time T, which causes the forecast at time T + 1 to explode and skew the overall average forecast error. This problem did not occur for the other forecasting models, which did a better job smoothing out the random noise.

Table 5
Disclosure risks and VAR model accuracy for the unprotected, k-nTS+ ($k = 3, M = 1.5$), and VAR-based approaches applied to the original M3 data. The * values indicate a disclosure risk value computed from the privacy analysis on the full data, i.e., the average $\bar{P}$ if an adversary collected 10 lagged values over time.

Privacy method	Shared data	Shared model	$\bar{P}$	$\bar{P}^f$	Accuracy (% change MAE)
Unprotected	Full data	No	98.4%	18.4%	0%
k-nTS+ (k = 3, M = 1.5)	Full data	No	8.2%	4.3%	107.7%
AN (s = 0.25)	Lags only	Yes (VAR)	*71.1%	8.6%	111.2%
k-nTS+ (k = 3, M = 1.5)	Lags only	Yes (VAR)	*8.2%	3.7%	>1000%
VAR-simulated	Lags only	Yes (VAR)	9.80%	18.0%	21.70%

improvement in forecast accuracy. The feature selection method in k-nTS+ results in approximately 2%–3% improved accuracy over k-nTS, albeit with a slight increase in the privacy risk measures.

Additional accuracy results for k-nTS+ ($k = 3, M = 1.5$) protected original data and k-nTS+ ($k = 3$) protected rate data, broken down by forecasting model and M3 data subset, are provided in Tables 7 and 8 of the appendix. Table 7 shows that TES had the best accuracy on the original data, but was one of the least accurate for the rate data. The LGBM and RNN models performed well on both unprotected and protected rate data, while auto-ARIMA excelled in most cases except for k-nTS+ ($k = 3, M = 1.5$) protected original data. While not displayed in the table, the transformed forecasts based on unprotected rate data for LGBM and RNN models show notable accuracy improvements of 14.6% and 10.0%, respectively, over forecasting the unprotected original series, which aligns with established practices in time series normalization for global forecasting models (Hewamalage et al., 2021, 2022). Table 8 shows that the original 'Monthly Micro' data had an acceptable increase in forecast error of 13.8%, but all other subsets had increases in forecast error of more than 65%. Notably, the rate forecasts and transformed rate forecasts had acceptable changes in accuracy on most M3 subsets, with the exception of 'Monthly Finance' and 'Monthly Micro'.

4.2.3. Can we share model parameters instead of protected data?

We apply the VAR-simulated approach discussed in Section 3.5 and attempt to identify the simulated series using the privacy analysis in Section 3.6. Rather than sharing simulated values with the forecaster, another option is for the data owner to share protected lagged values, which we assess using the best-performing k-nTS+ protected original and rate data sets and the additive noise ($s = 0.25$) data sets. We compare the accuracy and privacy of these parameter-sharing approaches to forecasting using VAR models trained on k-nTS+ protected original and rate data sets.

Table 5 reveals that while the VAR-simulated approach yields the best forecast accuracy (22% reduction over unprotected data) with acceptable identification disclosure risk, it barely reduces the forecast disclosure risk relative to the unprotected data. In other words, the time series are identifiable based on forecasts from VAR model parameters and simulated lagged data. This confirms the results from the simulation study in Section 4.1; on the original time series scale, good forecasts are not private.

The results in Table 6 show that the rate transformation enables private, accurate forecasts *because the forecasts are highly similar across all series*. Even on the unprotected data, just 2.5% of forecasts can correctly identify a time series. The disclosure risks for both the full k-nTS+ protected data sets and the VAR-simulated approach are acceptably low, while the best accuracy is achieved on the full k-nTS+ protected data sets for both the rate and transformed rate forecasts. Overall, the VAR-based approaches that utilize protected lags performed poorly. Even with weak privacy using additive noise-protected lags ($s = 0.25$), forecast accuracy is about 3% worse on the original data and 44% worse on the rate data relative to k-nTS+.

4.2.4. Time series feature selection

Fig. 5 summarizes the results of the feature selection method by M3 data subset and feature. The uncolored coordinates indicate either that a feature had RReliefF weight $\omega_n \leq 0$ and was eliminated in the first feature selection step, or it could not be computed for a given data set, such as the seasonal features for the yearly data. Surprisingly, some features that were found to be important in the literature review in Section 2, such as entropy and seasonal strength, were rarely selected, which implied that they were generally not the most useful for maintaining the forecast accuracy of protected data. The benefit of the proposed feature selection methodology becomes apparent when considering the small proportion of blue coordinates in each column, indicating that only a few features are needed to obtain the best accuracy when predicting the error of forecasting models. In Section 7 of the appendix, we include additional figures showing the RReliefF weights across all features and the MAE of the random forest predictions of the MAE of all forecasting models for both the original and rate data sets. The features that had the largest average RReliefF weights, such as spike and max variance shift, tended to be selected most frequently regardless of the data under consideration. Some features such as curvature and linearity had large weights but were not as frequently selected by the RFE stage on the original data. However, these features were selected much more frequently for the rate data. Others, such as trend, Hurst, and X ACF, were selected relatively frequently for the original data, but were eliminated much more frequently for the rate data.

Table 6
Disclosure risks and VAR model accuracy for the unprotected, k-nTS+ ($k = 3$), and VAR-based approaches applied to the M3 rate data. The * values indicate a disclosure risk value computed from the privacy analysis on the full data, i.e., the average $\bar{P}$ if an adversary collected 10 lagged values over time.

Privacy method	Shared data	Shared model	$\bar{P}$	$\bar{P}^f$	Accuracy (% change MAE)	Accuracy (% change MAE) original scale
Unprotected	Full data	No	98.4%	2.5%	0%	0%
k-nTS+ (k = 3)	Full data	No	7.8%	1.5%	−4.4%	42.3%
AN (s = 0.25)	Lags only	Yes (VAR)	*91.3%	1.5%	39.6%	227.1%
k-nTS+ (k = 3)	Lags only	Yes (VAR)	*7.8%	2.2%	3.9%	613.1%
VAR-simulated	Lags only	Yes (VAR)	2.5%	1.8%	−0.7%	110.7%

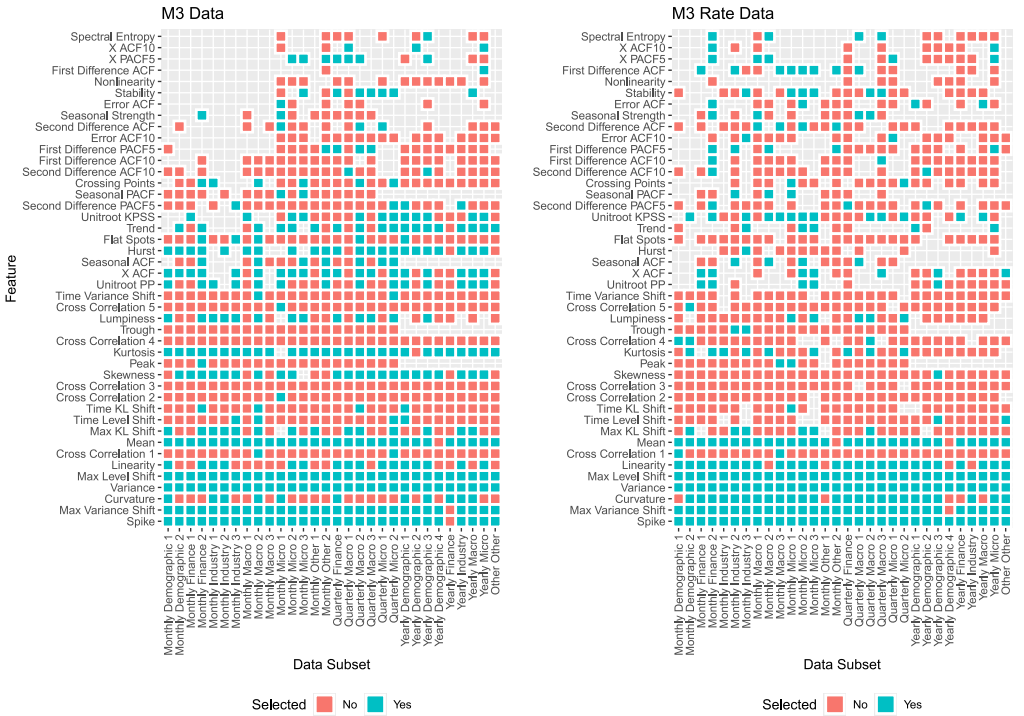


Fig. 5. Results of the proposed feature selection methodology. Blue (red) coordinates indicate that a given feature was (was not) selected to be used in k-nTS+ for the corresponding data set. Gray coordinates indicate that either a feature could not be computed for a given data set (e.g., seasonal features for yearly data) or a feature had an RReliefF weight $\omega_n \leq 0$ and was eliminated based on the first stage of the feature selection methodology. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.2.5. Time series features after protection: A tale of two (protected) time series

Fig. 6 illustrates the advantages of the k-nTS+ privacy method using two time series from the original M3 'Monthly Micro' data (where the tradeoff between privacy and forecast accuracy was acceptable) with desirable (blue circles) and undesirable (red triangles) features. Plot (i) shows the unprotected series, plot (ii) illustrates the results using k-nTS+ with $k = 3$ and $M = 1.5$, and plot (iii) illustrates the results using additive noise with $s = 1.5$. For the k-nTS+ protected series, there are only minor visual changes. For additive noise, there are drastic changes to both series. Table 11 of the appendix displays the values of several time series features before and after protection and shows that the low spectral entropy and high Hurst coefficient values of the desirable time series indicate good forecastability.

Fig. 7 displays boxplots of the values of the time series features that were selected in every subset of the M3 'Monthly Micro' data set, which included 474 time series. The feature values from the original M3 series are compared to the values from the series simulated from the VAR models and the protected series from the k-nTS+ ($k = 3, M = 1.5$), k-nTS ($k = 3$), AN ($s = 1.5$), and DP ($\epsilon = 4.6$) privacy methods. Random noise-based privacy methods significantly change the distributions of most features, leading to poor forecast accuracy. On the other hand, the k-nTS and k-nTS+ methods better preserve the feature distributions, while the series simulated from the VAR models in some cases exhibit feature values which are better for forecasting (e.g., large values of X ACF) but are unrepresentative of the features observed in the unprotected data. Compared to the distributions from the

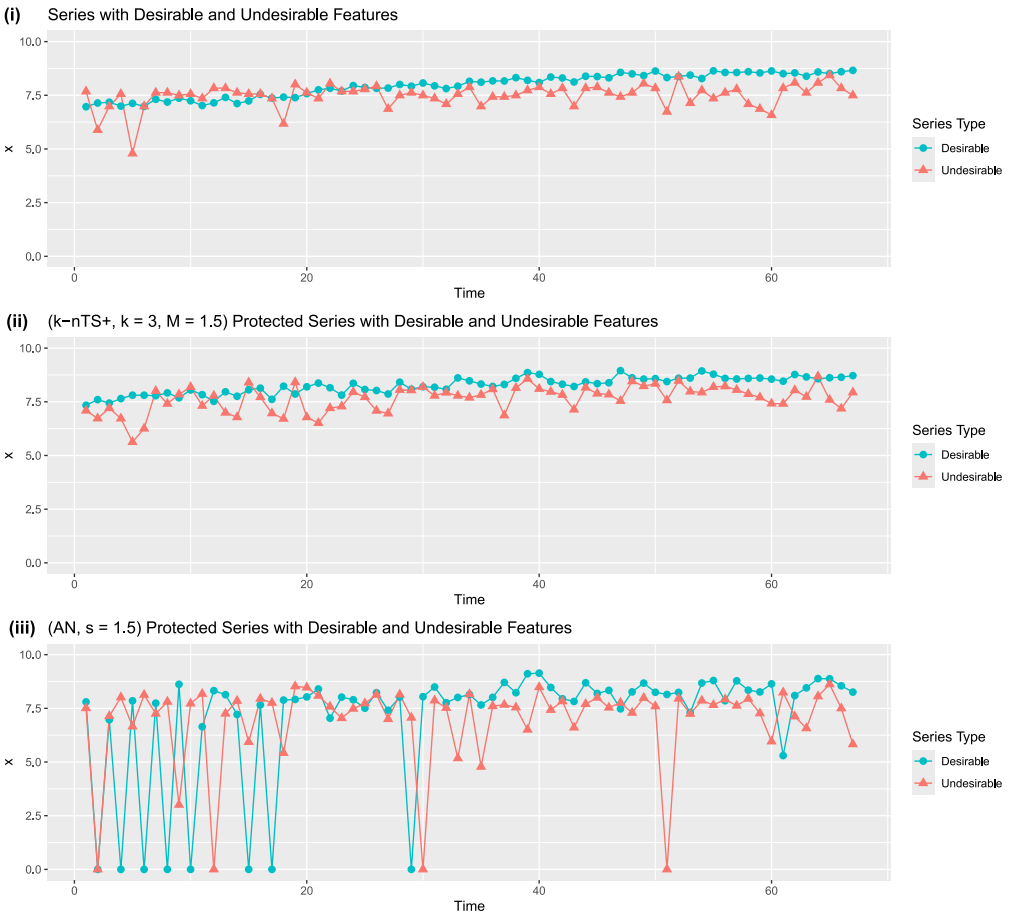


Fig. 6. A tale of two (protected) time series: Comparison of unprotected, k-nTS+ ($k = 3, M = 1.5$), and AN ($s = 1.5$) protected versions of original 'Monthly Micro' series with desirable (blue dots) and undesirable (red triangles) features. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

k-nTS+ protected series, the distributions from the VAR-simulated series are less similar to the distributions from the unprotected series. Overall, the features of the k-nTS+ protected series are most similar to those of the unprotected series, which resulted in the best forecast accuracy of the privacy methods we considered.

Table 12 of the appendix displays the mean absolute difference in cross-series correlations between the original unprotected and protected M3 data sets. Of the protected series with acceptable privacy, the bounded k-nTS+ ($k = 3, M = 1.5$) protected series produced the smallest mean absolute difference in cross-series correlations (approximately 0.21). By comparison, the VAR-simulated series had a mean absolute difference of 0.45. The distributions of the cross-correlations for the unprotected, AN ($s = 1.5$), DP ($\epsilon = 4.6$), k-nTS+ ($k = 3, M = 1.5$), k-nTS ($k = 3$), and VAR-simulated protected series are shown in Fig. 4 of the appendix. The VAR-simulated series produced the largest distortion in the distribution of cross-correlations, with 50% of the correlations being approximately equal to one. The other privacy methods reduced the strength of cross-correlations towards zero, with the k-nTS+ ($k = 3, M = 1.5$) cross-correlation distribution being most similar to that of the unprotected series.

We use principal components analysis (PCA) to analyze the joint distribution of the features that were selected for the majority of the original M3 data subsets (at least 28/30 cases), namely, variance, max level shift, spike, max variance shift, mean, and kurtosis (Kang et al., 2017; Spiliotis et al., 2020). Let $\mathbf{C}$ denote the matrix containing the combined features from all unprotected and protected series from the methods examined in Fig. 7. The principal components are calculated as

$$\begin{bmatrix} \text{PC1} \\ \text{PC2} \end{bmatrix} = \begin{bmatrix} 0.49 & 0.50 & 0.30 & 0.48 & -0.40 & 0.15 \\ -0.15 & 0.05 & -0.29 & 0.19 & 0.23 & 0.89 \end{bmatrix} \mathbf{C}, \tag{8}$$

where PC1 and PC2 denote the first two principal components, which are linear combinations of the six time series feature values. The smaller the distance between two points in the principal components space, the more similar the corresponding time series are in terms of the six features. The first two principal components explain 78% of the variation in the time series features. The first component increases with the values of features related

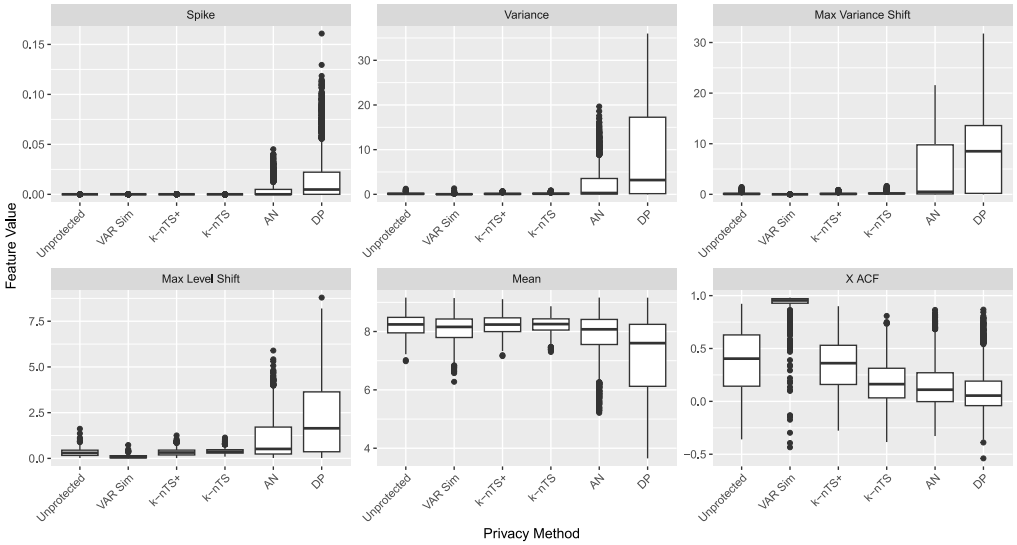


Fig. 7. Distributions of the time series features selected in every M3 'Monthly Micro' subset under the VAR-simulated, k-nTS+ ($k = 3, M = 1.5$), k-nTS ($k = 3$), AN ($s = 1.5$), and DP ($\epsilon = 4.6$) privacy methods.

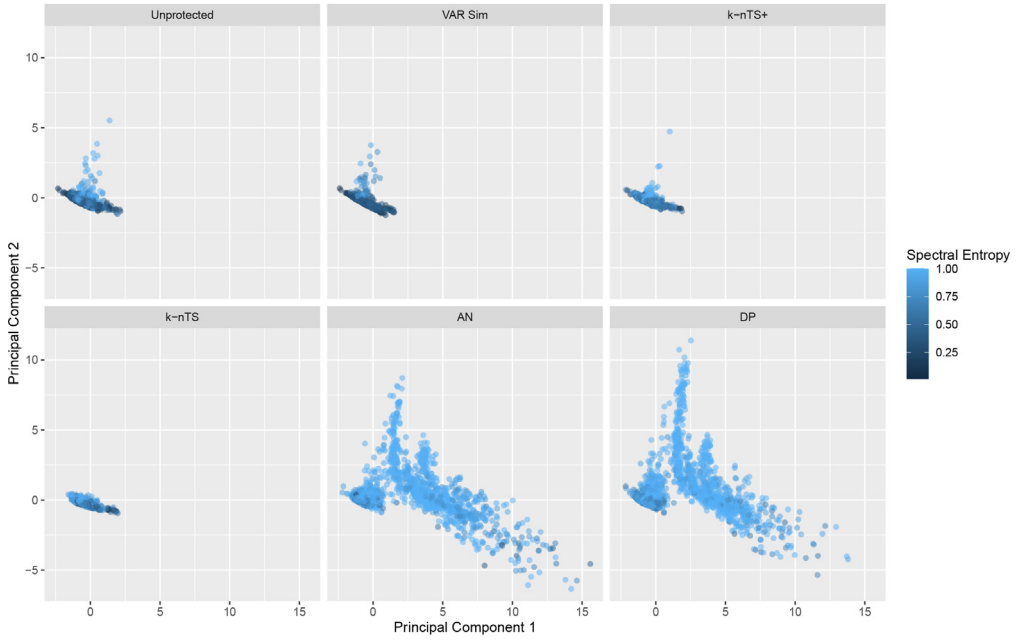


Fig. 8. Principal components calculated from the most frequently selected features across all original M3 time series.

to the variance, spikiness, and changes in the mean of the time series, and decreases with the mean of the time series. The second component is largely determined by kurtosis, with large positive values of kurtosis increasing the values of PC2, and vice versa.

Fig. 8 compares the distribution of principal component values for all of the unprotected original M3 time series and the protected series under each privacy method. Visually, the features from the k-nTS+ protected data and the VAR-simulated series preserve most of the unprotected feature space. Many of the series protected using differential privacy and additive noise contain combinations of features that do not occur in the unprotected data, such as larger (smaller) values of PC1 (PC2) corresponding to large values of variance, max variance shift, spike, and max level shift shown in Fig. 7. There are also many series that have similar values of PC1 but have a large range of PC2 values in the positive direction, which correspond to increases in kurtosis which indicate a series with a peak near the mean and heavy tails.

Fig. 8 also depicts the spectral entropy of the time series using the color of the points on the principal component axes. The poor forecasting performance of the series protected using additive noise and differential privacy is reflected by the light blue color of many points, indicating high values of spectral entropy. On the other hand, the VAR-simulated series tend to have lower spectral entropy than the unprotected series, even though these series had reduced forecast accuracy (i.e., the signal contained in the VAR-simulated series is not as predictive of future unprotected values). Furthermore, while the distribution of features that are most important for maintaining forecast accuracy is well maintained in the *k*-nTS+ protected series, forecast accuracy using the *k*-nTS+ protected data still decreased on average, and the average spectral entropy increased from 0.50 in the unprotected series to 0.70 in the *k*-nTS+ protected series. This is consistent with the results in Spiliotis et al. (2020) that showed using multiple linear regression that an increase in spectral entropy (or randomness) of a time series is associated with an increase in forecast error when holding many other time series features constant. In Fig. 5 of the appendix, we plot the first two principal components for the unprotected, VAR-simulated, and *k*-nTS+ ($k = 3, M = 1.5$) series recalculated to include all features[7] and find that *k*-nTS+ preserves the most accurate representation of the overall feature distribution.

4.2.6. Computational cost

The analysis of the M3 data utilized a desktop PC with a Ryzen 7900x CPU, 64 GB DDR5 6000MT/s CL30 RAM, and an RTX 4090. Fig. 9 (left) illustrates the total time (in seconds) to perform feature selection and swapping when generating the *k*-nTS+ protected M3 data subsets as a function of the total number of time periods swapped. The plot shows a roughly linear increase in computational cost, averaging 0.03 additional seconds for one additional swapped period (adjusted R^2 of 0.66). The RReliefF and RFE steps in the feature selection process respectively increase computation time by 0.07 and 2.85 s (adjusted R^2 values of 0.91 and 0.92) on average for one additional time series. On average, *k*-nTS swapping sees an increase of 0.58 s for one additional time series, holding the time series length constant, and an increase of 1.36 s when the average time series length increases by one, holding the number of time series constant (adjusted R^2 value of 0.90). The *k*-nTS+ process for the most computationally intensive subset, 'Monthly Micro', took 1670 s (just under 28 min).

Fig. 9 (right) illustrates the computation time in seconds for generating *k*-nTS+ protected versions of the 'Monthly Micro' time series, broken out by protection process steps. The majority of computation time was dedicated to the RFE step (about 1300 s). All of the results in this section are based on the 44 time series features, including the features calculated using the 'tsfeatures' R package and the cross-correlation features. Computation time will increase (decrease) with an increase (decrease) in the number of features input into the initial feature selection process.

[7] We exclude three features that capture the index of the maximum shifts in the level, variance, and Kulback–Leibler divergence, since the VAR-simulated series have fewer time periods overall. We include the absolute value of the cross-correlation features (eigenvectors corresponding to the first five principal components of the cross-series correlation matrix), since the signs of the eigenvector values are arbitrary. In addition, see Table 12 and Fig. 4 for analysis of the changes in time series cross-correlations.

5. Conclusion

The answer to the question 'Can we protect time series data while maintaining accurate forecasts?' is: It depends on the context. In order to achieve an acceptable privacy–forecast accuracy tradeoff, protected past values/features had to be dissimilar from unprotected values/features to preserve privacy, while enabling forecasts that were as similar as possible to future values. This tension prevented any of the privacy methods we studied from consistently maintaining a usable level of forecast accuracy. These findings are consistent with past research (Gonçalves et al., 2021; Nin & Torra, 2009). We also found that sharing model parameters enables accurate forecasts that can be used to reidentify time series.

To achieve an acceptable privacy–forecast accuracy tradeoff, we transformed time series into rates, which increased the similarity of values/features. This transformation enabled our proposed method to produce protected data with a small reduction in forecast accuracy. Compared to a differentially private method, our method improved forecast accuracy substantially, with comparable disclosure risks.

For limitations, the identification and forecast disclosure risk metrics could be considered ad hoc, as they do not guarantee differential privacy. However, they are more readily interpretable and reasonable according to the GDPR (Bale et al., 2024). We could have considered other privacy metrics, such as the risk of reconstruction attacks (Gonçalves et al., 2021; Goncalves et al., 2021) which would preclude the sharing of unprotected lagged time series values and model parameters. Reconstruction attacks are unlikely to succeed against *k*-nTS+ protected data, due to the randomness of the swapping process. Differentially private time series are immune to reconstruction attacks (in theory), but forecasts from differentially private data at acceptable levels of $\epsilon \leq ln\,3$ were unusable in this paper. Increasing ϵ improved accuracy but resulted in high disclosure risks. Overall, the differentially private time series in our application were either not useful for forecasting or not protective against identification and forecast disclosure.

Future work could extend the *k*-nTS+ method to decentralized scenarios where data owners could apply the method to their own time series before sharing/pooling data together. In cases of limited data, data owners could consider swapping values with synthetic time series with desirable features (Kang et al., 2020). Our method could also be used by a trusted central party to protect data from multiple data owners before sharing the combined protected data with participating parties. This could produce interesting opportunities for forecasting with global

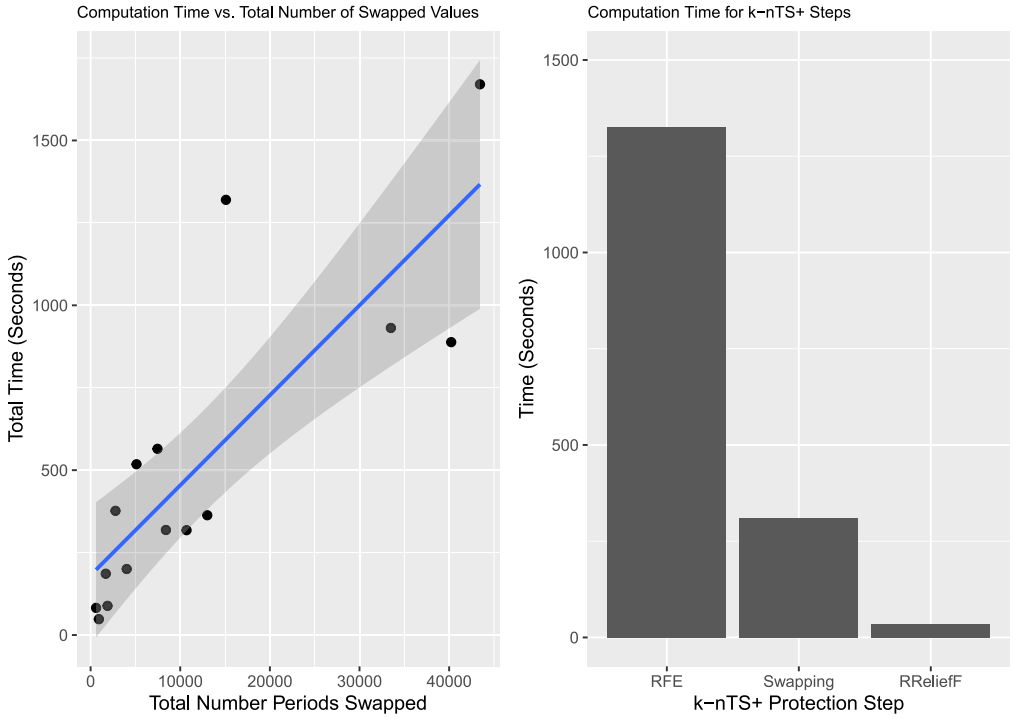


Fig. 9. Total computation time (in seconds) to create the *k*-nTS+ protected data sets (left plot) and a breakdown of the computation time (in seconds) of steps in the protection process (right plot).

models trained on a combination of a data owner's unprotected data and protected data from other parties. This approach could increase the willingness of owners to share data by alleviating privacy concerns with storing sensitive data in a central location (Goncalves et al., 2020).

It will also be important to define additional privacy attacks against time series data. As discussed by Reiter (2023) in the context of synthetic data, protection against identification disclosure may be unnecessary in some contexts. Instead, only protecting against attribute disclosure (when an adversary discerns sensitive information about a time series) may enable the sharing of protected time series with better forecast accuracy. Future work could also examine additional forecasting strategies for protected data, such as forecast combinations or using time series features to predict the best-performing forecasting model(s) for a protected data set.

CRediT authorship contribution statement

Cameron D. Bale: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Matthew J. Schneider:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Jinwook Lee:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.01.001.

Data and code availability

All material (data and code) necessary to reproduce the results presented in this manuscript is available as part of the reproducibility package at github.com/cdbale/Privacy-and-Forecasting-Empirical.

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